

ZIPENG HU

Curriculum Vitae

Kavli Institute for Astronomy & Astrophysics
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Employment

2025–present **Boya Fellow, The Kavli Institute for Astronomy and Astrophysics, Peking University, Beijing.**

Education

2021–2024 **Ph.D., Astronomy and Astrophysics, Australian National University, Canberra.**

2018–2020 **M.S., Astronomy and Astrophysics, Australian National University, Canberra.**

2014–2018 **B.S., Physics, The Chinese University of Hong Kong, HKSAR.**

Publications

Summary: 5 first-author papers, 1 leading-author paper, 7 publications in total.

(Notes: * – corresponding author)

First/Corresponding-author papers

- 2026 **Zipeng Hu***, Ke Wang, Mark R. Krumholz, and Keyun Su. A Pan-galaxy Study of Synthetic Giant Molecular Filaments: A Turbulence-dominated Life Cycle. *ApJS*, volume 282, page 34, February 2026.
- 2025 Chuhan Zhang, Zefeng Li*, **Zipeng Hu***, and Mark R. Krumholz. Understanding the mechanisms behind the distribution of galactic metals. *MNRAS*, volume 540, pages 3906–3918, July 2025.
- 2024 **Zipeng Hu***, Benjamin D. Wibking, Mark R. Krumholz, and Christoph Federrath. On the limitations of $H\alpha$ luminosity as a star formation tracer in spatially resolved observations. *MNRAS*, volume 534, pages 2426–2437, November 2024.
- 2023 **Zipeng Hu***, Benjamin D. Wibking, and Mark R. Krumholz. The sub-critical illusion: synthetic Zeeman effect observations from galactic zoom-in simulations. *MNRAS*, volume 521, pages 5604–5615, June 2023.
- 2022 **Zipeng Hu***, Mark R. Krumholz, Riway Pokhrel, and Robert A. Gutermuth. High-precision star-formation efficiency measurements in nearby clouds. *MNRAS*, volume 511, pages 1431–1438, February 2022.
- 2021 **Zipeng Hu***, Mark R. Krumholz, Christoph Federrath, Riway Pokhrel, and Robert A. Gutermuth. Reconstructing three-dimensional densities from two-dimensional observations of molecular gas. *MNRAS*, volume 502, pages 5997–6009, April 2021.

Co-author papers

- 2025 Arunima Arunima, Mark R. Krumholz, Michael J. Ireland, Chuhan Zhang, and **Zipeng Hu**. Fundamental limits to orbit reconstruction due to non-conservation of stellar actions in a Milky Way-like simulation. *MNRAS*, volume 543, pages 358–374, October 2025.

Fellowships & Awards

- 2025–present **Boya Fellowship**, for postdoc in Peking University.
- 2018 **Graduation with the First Class Honours**, for B.S. degree at the Chinese University of Hong Kong.
- 2017 **Charles. K. Kao Scholarship**, for overseas student research internships.
- 2017 **C.N. Yang Scholarship**, for students with excellent academic performance.
- 2016 **Yasumoto International Exchange Scholarship**, for overseas exchange program.
- 2015 **Good Earth Foundation Scholarship**, for outstanding performance in extra-curricular activities.

Conferences & Talks

- 2026 **Contributed talk at Perspectives in Star Formation: Senior Insights, Young Visions**, Shanghai, China.
- 2025 **Contributed talk at the Annual Conference on Molecular Clouds and Star Formation**, E'meishan, China.
- 2024 **Contributed talk at the Annual Conference on Molecular Clouds and Star Formation**, Nanning, China.
- 2023 **Contributed talk at the Annual meeting of ASA**, Sydney, Australia.
- 2023 **Poster presentation at PPVII**, Kyoto, Japan.
- 2022 **Seminar talk at CUHK**, HKSAR, China.
- 2022 **Seminar talk at NAOC**, Beijing, China.

Grants & Projects

PI projects

- 2026-2028 **NSFC Youth Grant**: The cross-scale interplay between B-field and gravity: study on the B-field regulation in molecular cloud formation and evolution.

CO-I projects

- 2026-2029 **NSFC General Grant**: Initial Conditions for the Formation of Massive Stars and Star Clusters: A Scaled Study Based on Unified ALMA Observations
- 2026-2029 **NSFC General Grant**: Cross-Scale Self-Consistent Computational Research on Astrophysical Processes in the Inner Regions of Protoplanetary Disks
- 2025-2029 **National SKA Program of China**: Research on Frontier Issues of Star Formation in the Entire Galactic Plane
- 2025 **ALMA Cycle 12**: How does gas is assembled from filament to cores? An ACA large-mosaic survey of Galaxy-wide linear filaments.

Teaching & Mentoring

- 2025 **Science Talent Training Program**: Supervise high school student research project. Students win the Second Prize of Beijing Youth Science & Technology Innovation Competition.
- 2023-present Co-advisor of Chuan Zhang and A. Arunima, graduate students at Australian National University.

Referees

Dr. Mark Krumholz

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Astronomy & Astrophysics*
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